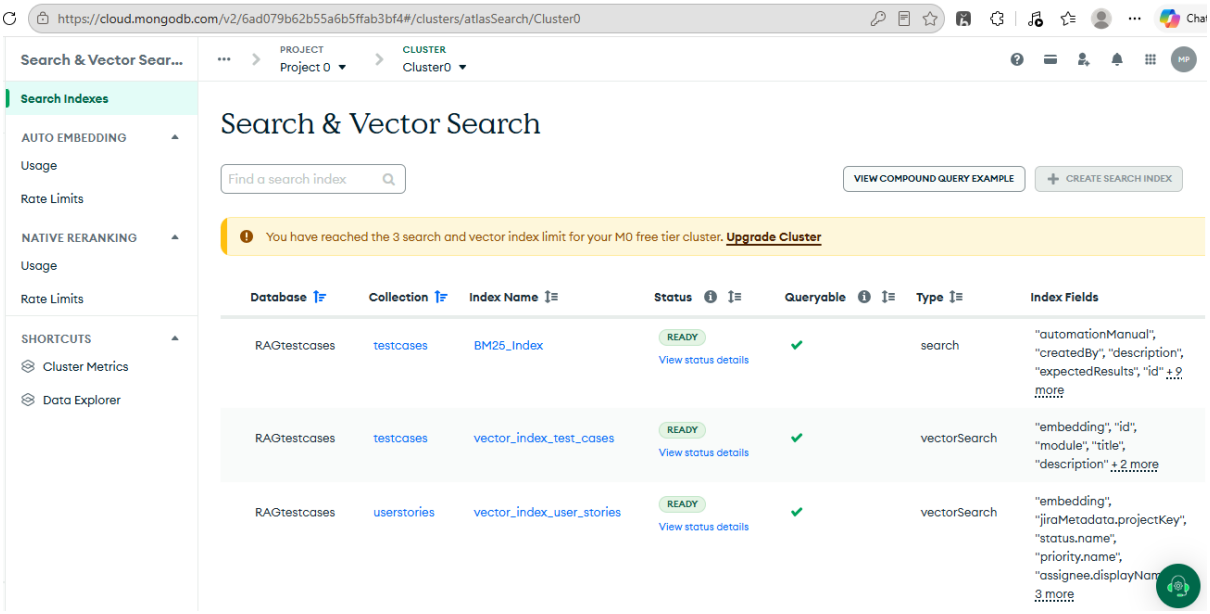
Home Assignment –

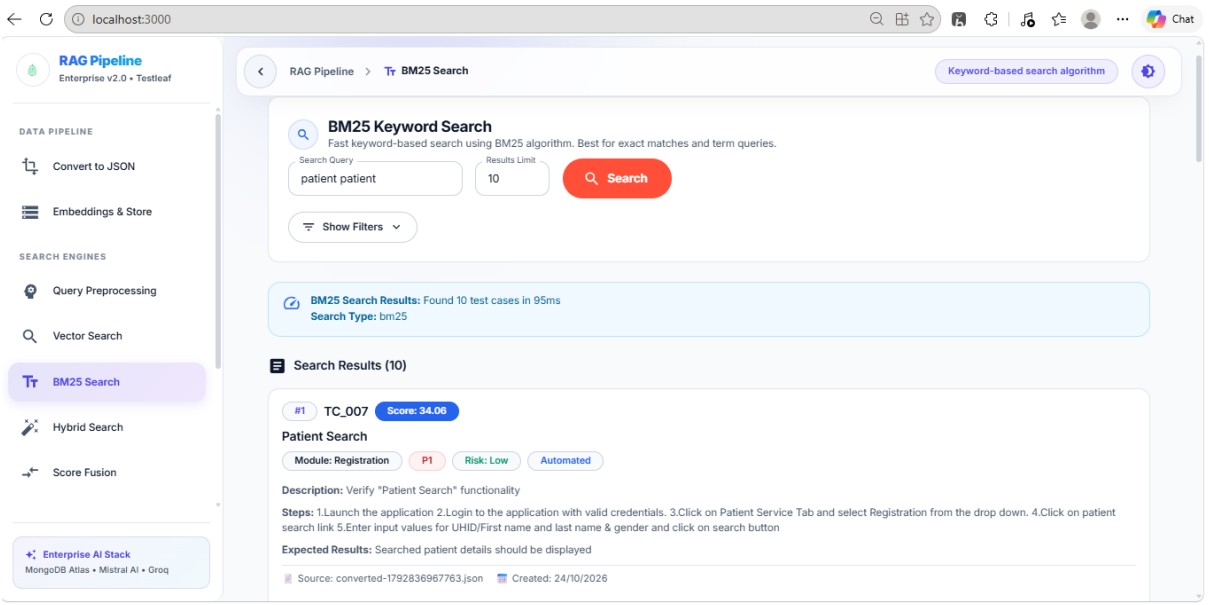
Create a BM25 index and explore till hybrid search.

Please refer the attached document for creating BM25 index.

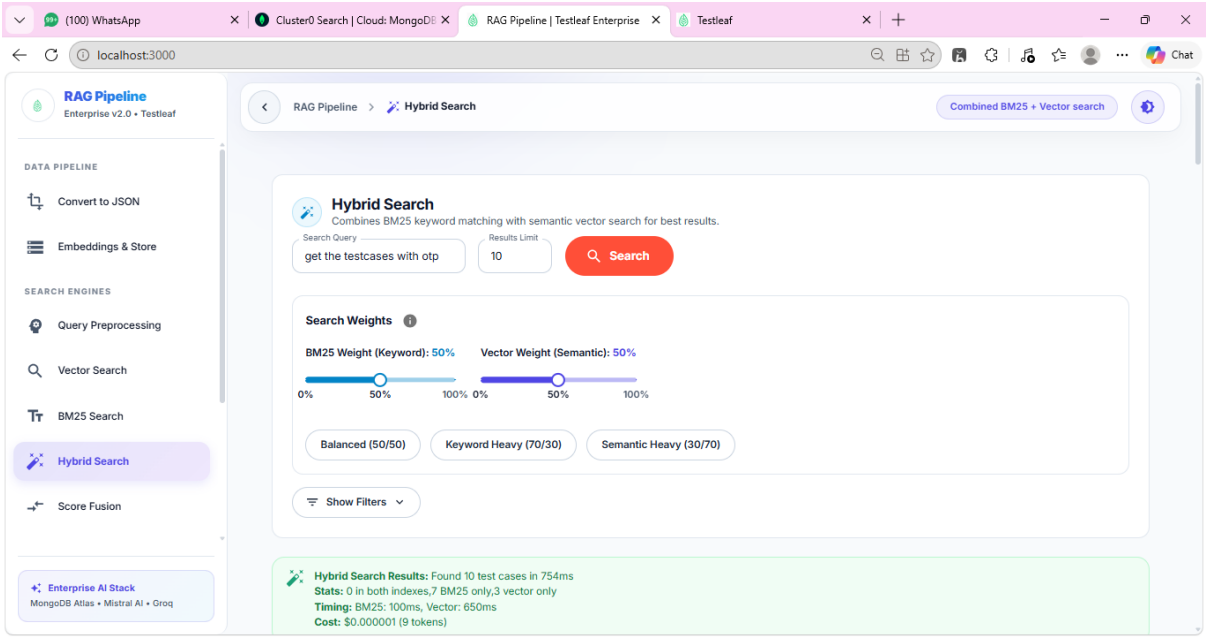
Created the BM25 Index in Mongo DB:

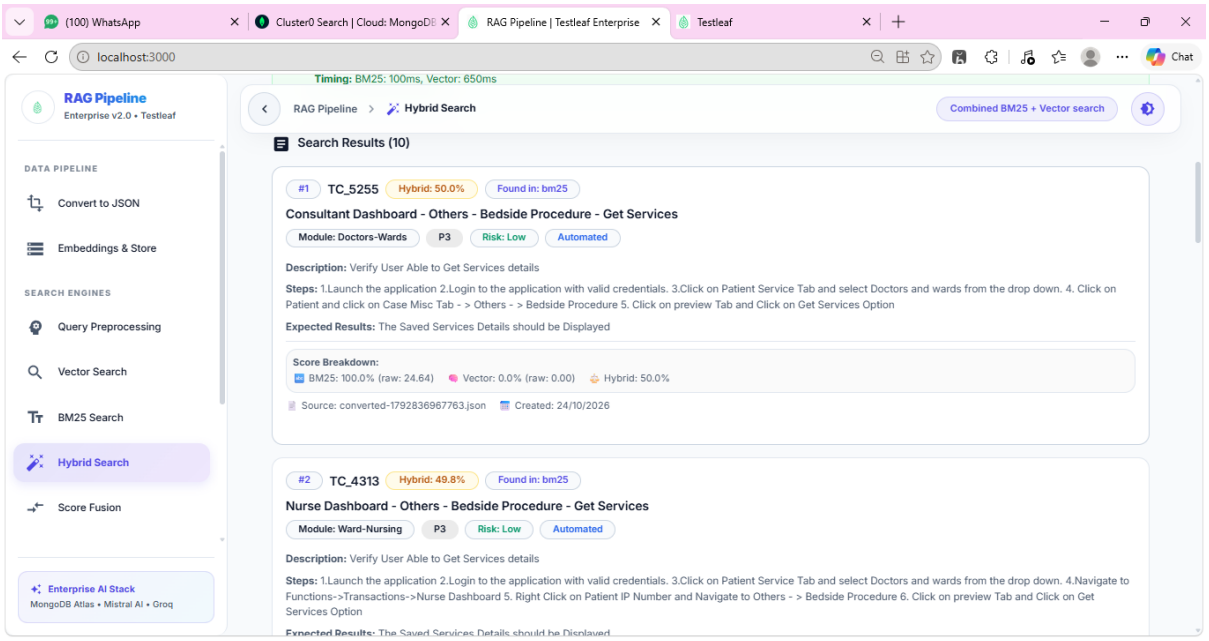


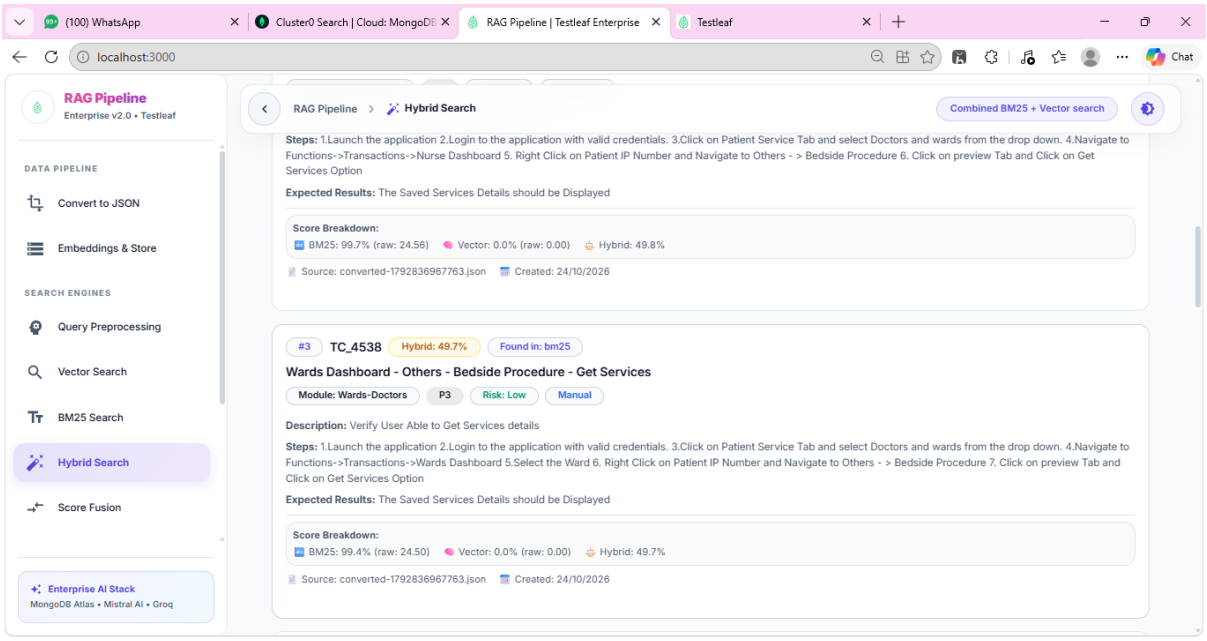
Able to do BM25 search:



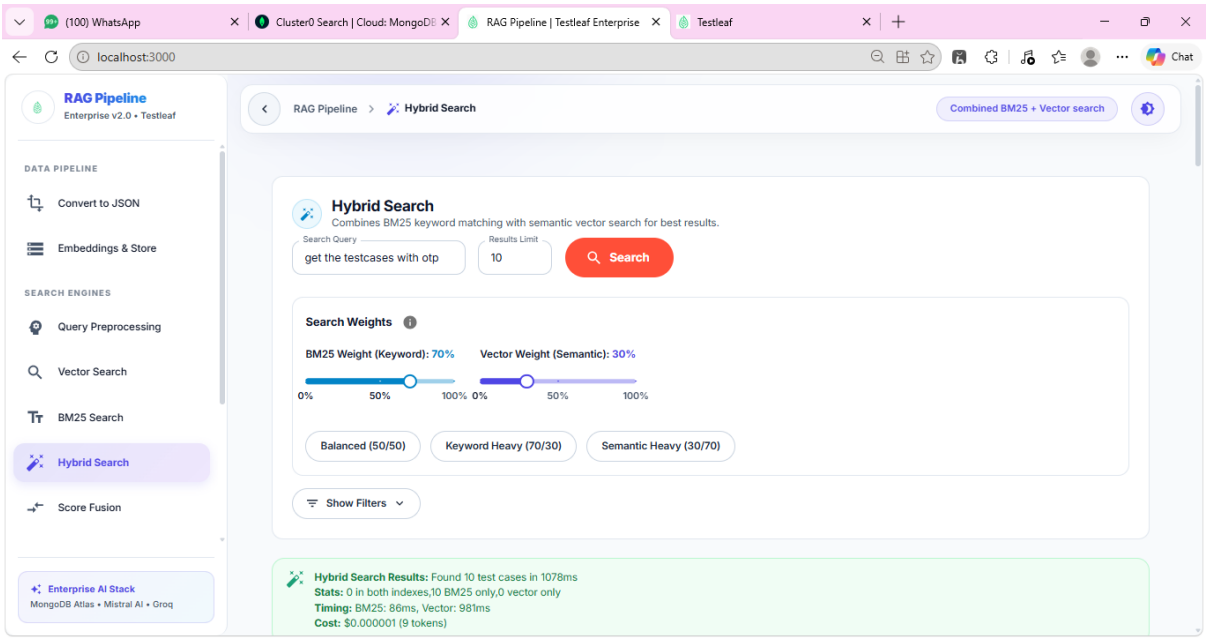
Hybrid search: Balance (BM25 – 50%, Vector – 50%)

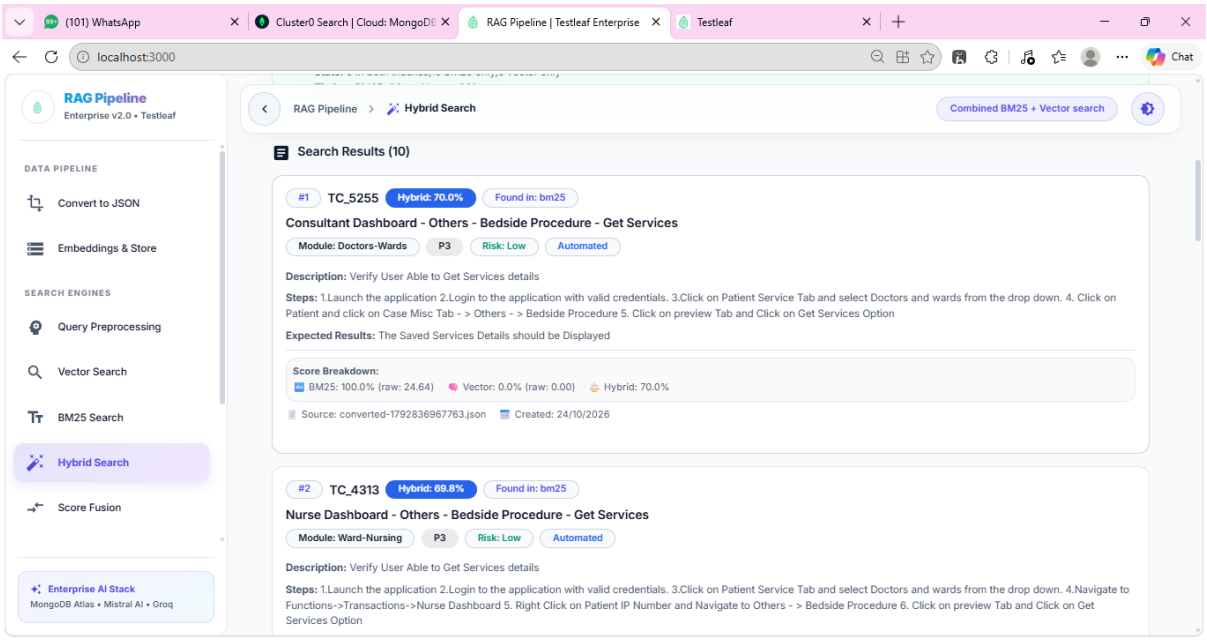


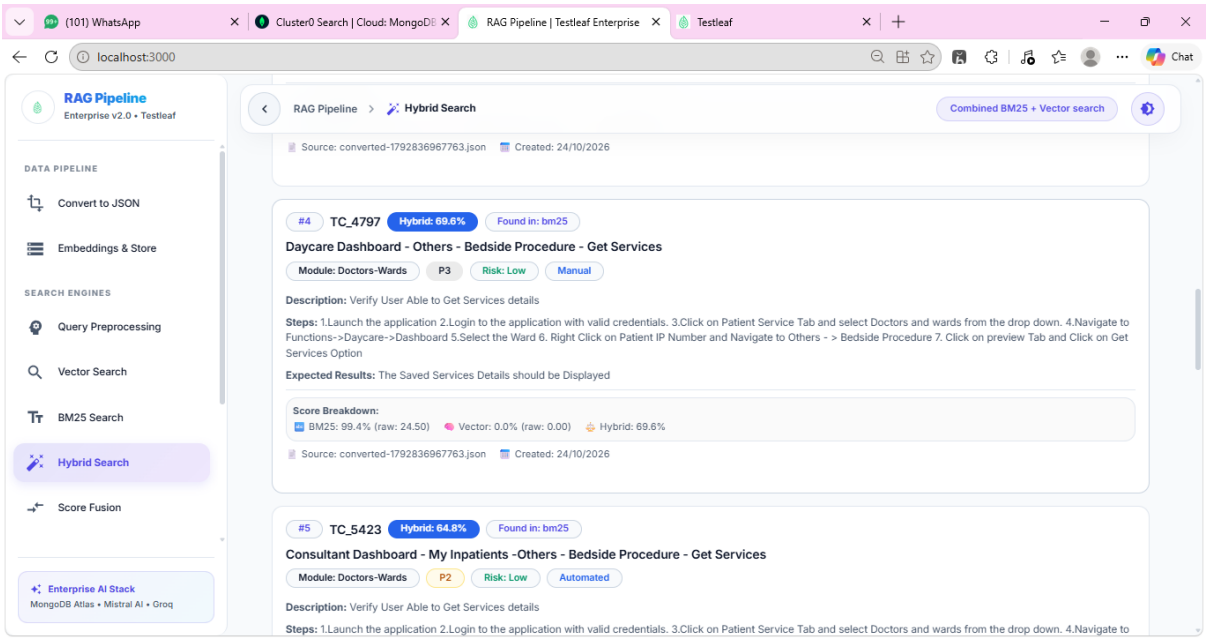




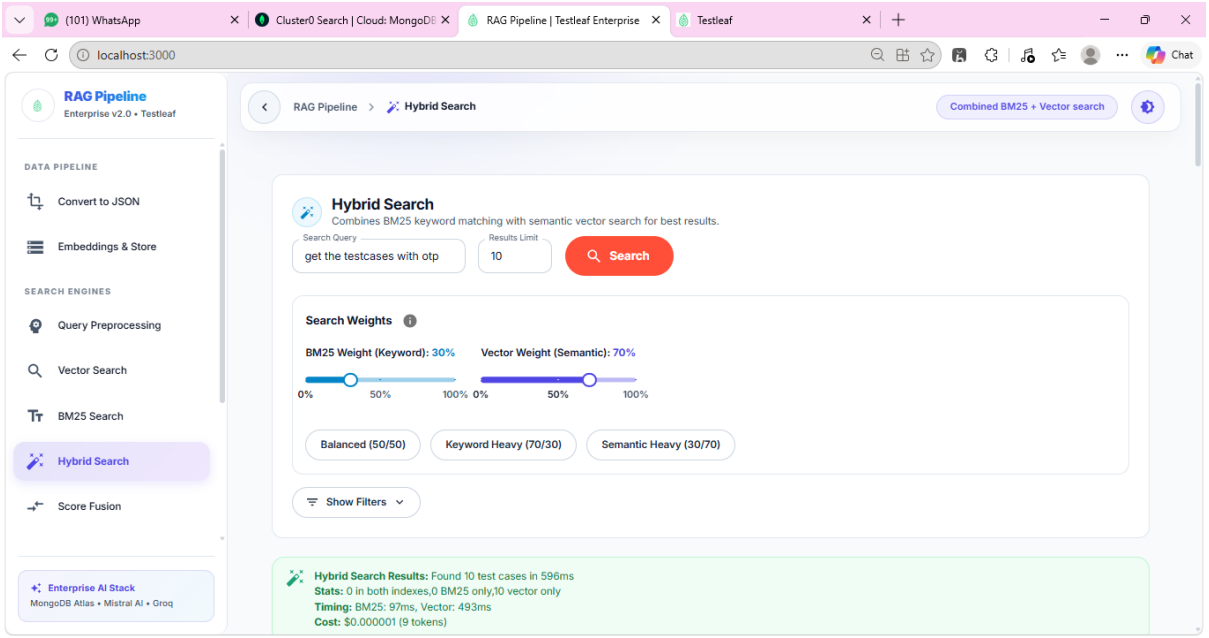
Keyword heavy search: (BM25-70%, Vector – 30%)

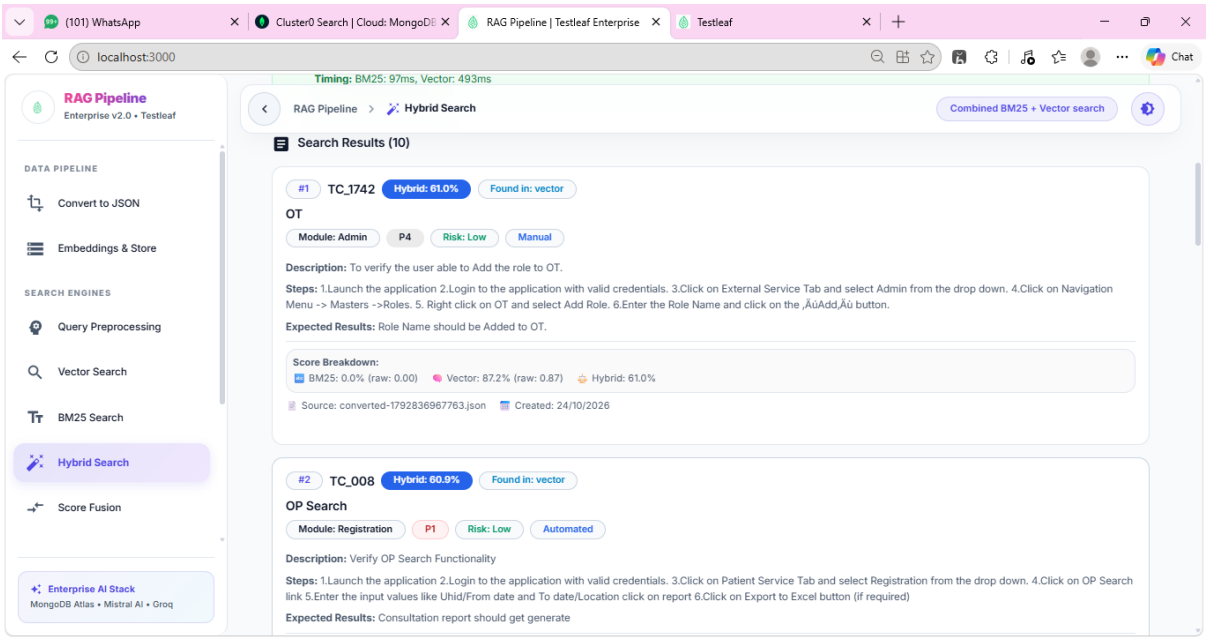


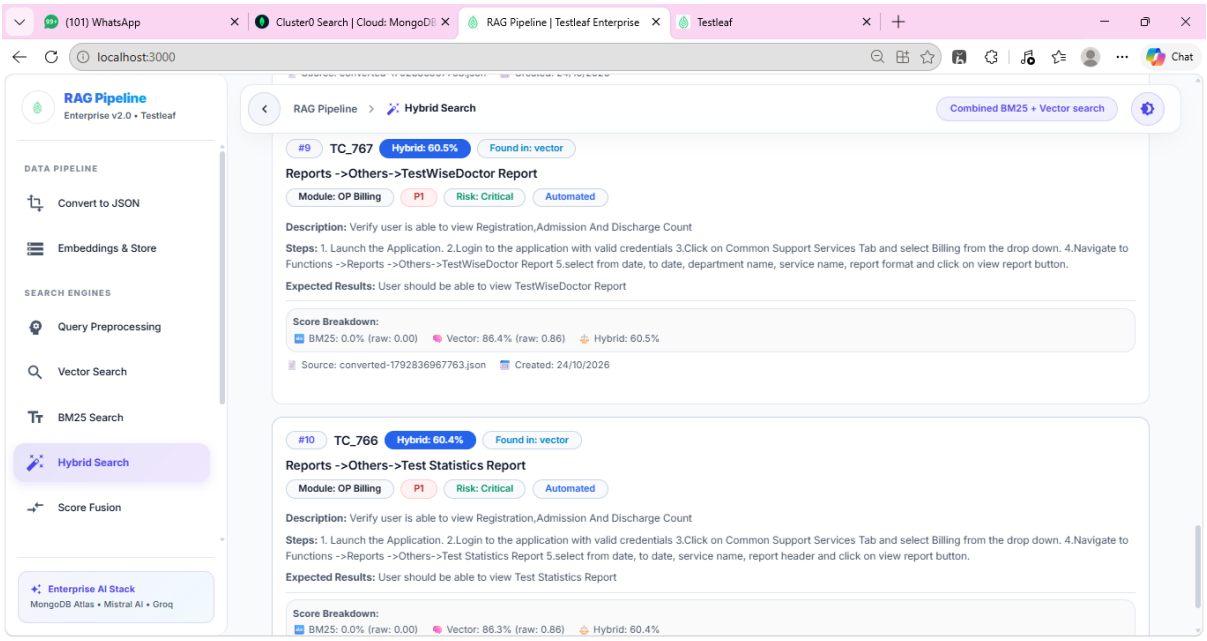

(101) WhatsApp
Cluster0 Search | Cloud: MongoDB
RAG Pipeline | Testleaf Enterprise
Testleaf
localhost:3000
Chat
RAG Pipeline
Enterprise v2.0 • Testleaf
DATA PIPELINE
Convert to JSON
Embeddings & Store
SEARCH ENGINES
Query Preprocessing
Vector Search
BM25 Search
Hybrid Search
Score Fusion
Enterprise AI Stack
MongoDB Atlas • Mistral AI • Groq
RAG Pipeline
Hybrid Search
Combined BM25 + Vector search
Search Results (10)
#1 TC_5255 Hybrid: 70.0% Found in: bm25
Consultant Dashboard - Others - Bedside Procedure - Get Services
Module: Doctors-Wards P3 Risk: Low Automated
Description: Verify User Able to Get Services details
Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on Patient Service Tab and select Doctors and wards from the drop down. 4. Click on Patient and click on Case Misc Tab - > Others - > Bedside Procedure 5. Click on preview Tab and Click on Get Services Option
Expected Results: The Saved Services Details should be Displayed
Score Breakdown:
BM25: 100.0% (raw: 24.64)
Vector: 0.0% (raw: 0.00)
Hybrid: 70.0%
Source: converted-1792836967763.json
Created: 24/10/2026
#2 TC_4313 Hybrid: 69.8% Found in: bm25
Nurse Dashboard - Others - Bedside Procedure - Get Services
Module: Ward-Nursing P3 Risk: Low Automated
Description: Verify User Able to Get Services details
Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on Patient Service Tab and select Doctors and wards from the drop down. 4.Navigate to Functions->Transactions->Nurse Dashboard 5. Right Click on Patient IP Number and Navigate to Others - > Bedside Procedure 6. Click on preview Tab and Click on Get Services Option


(101) WhatsApp
Cluster0 Search | Cloud: MongoDB
RAG Pipeline | Testleaf Enterprise
Testleaf
localhost:3000
Chat
RAG Pipeline
Enterprise v2.0 • Testleaf
DATA PIPELINE
Convert to JSON
Embeddings & Store
SEARCH ENGINES
Query Preprocessing
Vector Search
BM25 Search
Hybrid Search
Score Fusion
Enterprise AI Stack
MongoDB Atlas • Mistral AI • Groq
RAG Pipeline
Hybrid Search
Combined BM25 + Vector search
Source: converted-1792836967763.json
Created: 24/10/2026
#4 TC_4797 Hybrid: 69.6% Found in: bm25
Daycare Dashboard - Others - Bedside Procedure - Get Services
Module: Doctors-Wards P3 Risk: Low Manual
Description: Verify User Able to Get Services details
Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on Patient Service Tab and select Doctors and wards from the drop down. 4.Navigate to Functions->Daycare->Dashboard 5.Select the Ward 6. Right Click on Patient IP Number and Navigate to Others - > Bedside Procedure 7. Click on preview Tab and Click on Get Services Option
Expected Results: The Saved Services Details should be Displayed
Score Breakdown:
BM25: 99.4% (raw: 24.50)
Vector: 0.0% (raw: 0.00)
Hybrid: 69.6%
Source: converted-1792836967763.json
Created: 24/10/2026
#5 TC_5423 Hybrid: 64.8% Found in: bm25
Consultant Dashboard - My Inpatients -Others - Bedside Procedure - Get Services
Module: Doctors-Wards P2 Risk: Low Automated
Description: Verify User Able to Get Services details
Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on Patient Service Tab and select Doctors and wards from the drop down. 4.Navigate to

Semantic heavy search (BM25-30%, vector-70%)






(101) WhatsApp
Cluster0 Search | Cloud: MongoDB
RAG Pipeline | Testleaf Enterprise
Testleaf
localhost:3000
Chat
RAG Pipeline
Enterprise v2.0 • Testleaf
DATA PIPELINE
Convert to JSON
Embeddings & Store
SEARCH ENGINES
Query Preprocessing
Vector Search
BM25 Search
Hybrid Search
Score Fusion
Enterprise AI Stack
MongoDB Atlas • Mistral AI • Groq
RAG Pipeline
Hybrid Search
Combined BM25 + Vector search
#9
TC_767
Hybrid: 60.5%
Found in: vector
Reports ->Others->TestWiseDoctor Report
Module: OP Billing
P1
Risk: Critical
Automated
Description: Verify user is able to view Registration,Admission And Discharge Count
Steps: 1. Launch the Application. 2.Login to the application with valid credentials 3.Click on Common Support Services Tab and select Billing from the drop down. 4.Navigate to Functions ->Reports ->Others->TestWiseDoctor Report 5.select from date, to date, department name, service name, report format and click on view report button.
Expected Results: User should be able to view TestWiseDoctor Report
Score Breakdown:
BM25: 0.0% (raw: 0.00)
Vector: 86.4% (raw: 0.86)
Hybrid: 60.5%
Source: converted-1792836967763.json
Created: 24/10/2026
#10
TC_766
Hybrid: 60.4%
Found in: vector
Reports ->Others->Test Statistics Report
Module: OP Billing
P1
Risk: Critical
Automated
Description: Verify user is able to view Registration,Admission And Discharge Count
Steps: 1. Launch the Application. 2.Login to the application with valid credentials 3.Click on Common Support Services Tab and select Billing from the drop down. 4.Navigate to Functions ->Reports ->Others->Test Statistics Report 5.select from date, to date, service name, report header and click on view report button.
Expected Results: User should be able to view Test Statistics Report
Score Breakdown:
BM25: 0.0% (raw: 0.00)
Vector: 86.3% (raw: 0.86)
Hybrid: 60.4%